ROSALIN DASH

Full Stack .NET Engineer | AI Engineer

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PROFESSIONAL SUMMARY

Senior Full Stack .NET Engineer with **10+ years** of software engineering experience, including **5+ years** building enterprise .NET applications for **banking and insurance**. Skilled in **C#, .NET Core, ASP.NET Core Web API, SQL Server, and Angular/Vue.js**, with an **M.S. in Computer Science (Generative AI)**.

- **Backend & Cloud:** System design, SOLID, CQRS, microservices, Kafka, Polly, Azure Functions, GCP Vertex AI, Azure DevOps/GitHub Actions/Harness CI/CD, cloud migration.
- **AI / GenAI:** LLM multi-agent systems (LangChain, LangGraph), RAG, prompt engineering, GANs, GitHub Copilot.
- **Leadership & Quality:** Code reviews, mentoring, TDD with 95% code coverage, Agile/Scrum delivery.

TECHNICAL SKILLS

Languages	C#, TypeScript, JavaScript, Python, SQL, Java
AI / ML & GenAI	Generative AI, LLM Multi-Agent Systems, AI Agents, LangChain, LangGraph, RAG, Prompt Engineering, Guardrails, Embeddings, Vector Databases (Vertex AI), Synthetic Data Generation, GANs, Machine Learning, Deep Learning, Computer Vision, TensorFlow, GitHub Copilot
.NET / Backend	.NET Core, ASP.NET Core MVC & Web API, .NET Background Services, RESTful APIs, Microservices, Entity Framework, LINQ, Polly, Microsoft ILogger, System Design, Design Patterns (CQRS, Repository, DI), SOLID
Frontend	Angular, Vue.js, Vuex, React, TypeScript, HTML5, CSS3
Cloud & DevOps	Azure Functions, Azure SignalR Service, Azure AD, Service Bus, Application Insights, Azure Monitor, GCP Vertex AI, Cloud Migration, Docker, Kubernetes (OpenShift), Azure DevOps, GitHub Actions, Harness CI/CD, Git, GitHub, Apigee, Ansible
Security & Identity	SAML SSO, Azure AD, OAuth 2.0, JWT, Role-Based Access Control (RBAC), Guardrails
Testing & Process	xUnit, NUnit, Moq, Unit Testing, TDD, Playwright, ALM Octane, HyperExecute, Swagger/OpenAPI, Postman, JIRA, Agile/Scrum
Data, Messaging & Observability	SQL Server (Stored Procedures, Views, Partitioning), MongoDB, Apache Kafka, Avro, Confluent Schema Registry, Power BI, pandas, Splunk, KQL

PROFESSIONAL EXPERIENCE

Wells Fargo (via Source Code Technologies LLC), California, US **Aug 2024 – Present**

Tech Stack: *.NET Core 3.1/6, C#, TypeScript, Python, Angular, Microservices, CQRS, Kafka, Avro, Confluent Schema Registry, ASP.NET Core MVC, Polly, Microsoft ILogger, Splunk, Entity Framework, Docker, Harness, GitHub Actions, Git, xUnit, Moq, Playwright, ALM Octane, HyperExecute, GitHub Copilot, LangChain, LangGraph, Streamlit, GCP Vertex AI Vector Search, RAG, MongoDB, Apigee*

Senior Software Engineer

- Managed an **enterprise-level data center migration** of a **25-component microservices** application from **on-prem to SaaS cloud** platforms, including **Kafka producers/consumers, MongoDB, firewall rules**, and **Apigee** API gateway components; coordinated dependency mapping, migration planning, cutover, and post-migration validation across teams.
- Partnered with the **Apigee** proxy team on **ingress port mapping** and vendor **IP allowlisting** for the vendor application.
- Helped set up **SAML SSO** with **Azure AD** authentication for **role-based IAM** login, using QA-ENT accounts for non-prod and AD-ENT accounts for prod, and **JWT-based OAuth 2.0** tokens from Azure AD for secured API access.
- Developed the **Account Summary** module: a **.NET Core background service** that consumes third-party vendor events and publishes agreed **Kafka** events using **Avro** schemas via **Confluent Schema Registry**, and an **ASP.NET Core MVC & Web API** service exposing account data to consumers through **Apigee** proxies.
- Designed and built an **LLM multi-agent system** for **enterprise-level synthetic data generation**, building agents with **LangChain**, orchestrating them with **LangGraph**, and delivering a **Streamlit** front end; applied **prompt engineering** with **system prompts** and **guardrails** to sanitize user input, reduce **hallucinations**, and keep agents scoped to required outputs.
- Used **Python** with **GCP Vertex AI Vector Search**, **embeddings**, and **retrieval-augmented generation (RAG)** to align generated data to the closest **rule engine** match for **pre-condition (PREC) and post-condition (POSTC)** test data.
- Designed and developed a **configuration management** component in **.NET Core** that pulls the configuration baseline from a **Business Continuity Risk Management (BCRM)**-owned environment and publishes it across release tracks, **eliminating manual configuration deployments**; built with **Polly** retries, timeouts, and circuit breakers and covered by **xUnit/Moq unit tests**.
- Adopted a **configuration-as-code** approach by versioning configuration in **Git**, authoring a custom **GitHub Actions** workflow for the configuration repo where the enterprise standard workflow didn't fit, and using **GitHub Copilot** to summarize release-branch changes into **release summaries**.
- Integrated **Splunk** across microservice components and helped resolve **Splunk Observability Cloud** ingestion-limit overruns in non-prod by identifying gaps between component logging and the standards of the **enterprise team's log-handling system**, leading to a **change request (CR)** to update component code to standards-compliant logging.
- Helped drive the **ALM Octane** integration, guiding teams across components and making code changes to map **HyperExecute** test suites and case IDs to user stories and **Harness CI/CD** pipelines for **test traceability**.

- Collaborated with the automation team to establish **TypeScript Playwright** automation, identifying client-side traffic and server-side **proxy access** for the vendor application server; fixed **12 test scenarios** for **recovery module** automation.

Orion180 Insurance Services, Melbourne, FL **Dec 2020 – Mar 2024**

Tech Stack: *.NET Core, C#, Vue.js, Vuex, LINQ, SQL Server (Stored Procedures, Views), Python (pandas), Power BI, Azure Functions, SignalR, Application Insights, Azure Monitor, KQL, Azure DevOps CI/CD*

Software Engineer

- Automated multi-state **tax report generation** for the My180 web app using **C#, .NET Core, and Vue.js**, replacing a manual process and improving accuracy and **audit transparency** by flagging and fixing data errors before tax reporting.
- Automated **refund and check processing** for the Home Insurance web app by building **.NET Core RESTful APIs** in **C#** and a **Vue.js** interface, with a scheduled **end-of-day (EOD) batch process** that updates policy records in **SQL Server**, replacing a manual workflow for the finance team.
- Automated monthly financial reports, including **Premium Summaries and Commission Reports**, in **C#** using **LINQ** to aggregate policy and commission data from **SQL Server**, replacing manual report preparation for the finance team.
- Designed and built **Power BI dashboards** on **SQL Server stored procedures and views** for claims statistics and tax, check, and refund discrepancy tracking, giving finance first-time visibility to catch and resolve errors before reporting.
- Re-designed the **physical data model** for the **claims module** in **SQL Server**, implementing indexing strategies, constraints, **table partitioning**, and **SEQUENCE**-based claim numbering to improve data integrity and query performance.
- Identified, reported, and fixed **geographic risk data misalignment** by cleaning third-party location data with **Python (pandas)** and integrating it into **Power BI** reports, improving policy risk analysis and underwriting accuracy.
- Designed and adopted **Azure SignalR Service** with **Azure Functions** to enable faster, real-time **document processing**, pushing processing status to users instantly and improving batch throughput.
- Drove **performance improvements** by analyzing peak-hour usage and backend **API hit rates** with **Application Insights and KQL**, performing **root cause analysis**, reviewing data models, and introducing **physical data layer (PDL)** concepts; defined features for multiple teams.
- Led the **production support** team and created user stories for the team, **trained new hires** as full stack engineers, and **mentored and managed interns**.

Alten Calsoft Labs Pvt. Ltd., Bangalore, India **Aug 2016 – Aug 2018**

Tech Stack: *Python, Java, REST APIs, YANG, Network Management Systems, Linux, Git*

Senior Software Engineer

- Developed **Python** scripts to automate the extraction of network timing device data, streamlining the auto-registration process and reducing manual extraction time by **40%**.
- Deployed **Java** applications for device discovery and auto-registration, facilitating East-West communication through **REST API** calls, which enhanced network integration and scalability.
- Utilized **Java** to marshal and unmarshal device details, populating corresponding **YANG** modules to ensure accurate data representation and seamless integration within network management systems.

Hewlett Packard Enterprise, Bangalore, India **Sep 2015 – Aug 2016**

Tech Stack: *Python, Ansible, NFV, SNMP, CLI Development, Linux, Virtual Machines*

Technology Consultant I

- Designed and implemented a **Command-Line Interface (CLI)** tool that enabled users to perform operations on various **NFV** system components, improving user interaction and operational efficiency.
- Created **Ansible** playbooks to configure, deploy, and manage CLI tool virtual machines within the NFV system, ensuring consistent and rapid deployments.
- Developed **Python** scripts for end-to-end cleanup of NFV system hardware components, including **SNMP** configuration and status checks, enhancing system reliability and maintenance efficiency.

Tata Consultancy Services, Bangalore, India **Jan 2013 – Sep 2015**

Tech Stack: *Java, OpenDaylight (ODL), MD-SAL, YANG, RPC, Maven, SDN*

System Engineer

- Spearheaded the implementation of service layers using **MD-SAL** for the **OpenDaylight (ODL) Controller**, enhancing network programmability and management.
- Created **YANG** models and **Remote Procedure Call (RPC)** interfaces for efficient data management, facilitating seamless communication between network devices and controllers.
- Deployed **Maven** bundles as features in the ODL Controller, improving node flow management and client call handling, which resulted in more robust and scalable network operations.

EDUCATION

Master of Science (M.S.) in Computer Science, Generative AI Focus **Aug 2018 – Dec 2020**

Florida Institute of Technology, Melbourne, FL | GPA**: 3.75/4.0**

Thesis: Biometric Face Skin Tone Data Augmentation using a Generative Adversarial Network (Advisor: Dr. Michael C. King)

Coursework: Artificial Intelligence, Deep Learning, Biometric Authentication, Software Design Methodologies

Research Projects: GAN style transfer (**BeautyGAN, TensorFlow, Python**) to generate six Fitzpatrick skin-tone classes; **ArcFace** face recognition analysis on a 440-image makeup dataset.

Bachelor of Technology (B.Tech.) in Electronics and Communication **Aug 2008 – Jul 2012**

Gandhi Institute of Engineering and Technology, India | GPA**: 8.55/10**